Отчет по практической работе №2

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1. Информация о кластере

1.1 Статус Minikube

Проверка установки kubernetes

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$ kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$
```

Запуск minikube

```
$ minikube start --driver=docker --cpus=4 --memory=8192
😄  minikube v1.38.1 на Microsoft Windows 11 Home Single Language 25H2
✨  Используется драйвер docker на основе существующего профиля
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
💾  Скачивается Kubernetes v1.35.1 ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 3.72 Mi
    > index.docker.io/kicbase/sta...:  519.58 MiB / 519.58 MiB  100.00% 2.27 Mi
❗  minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downloaded docker.io/kicbase/stable:v0.0.50@
sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93 as a fallback image
🤷  docker "minikube" container is missing, will recreate.
🔥  Creating docker container (CPUs=4, Memory=8192MB) ...
🐳  Подготавливается Kubernetes v1.35.1 на Docker 29.2.1 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Компоненты Kubernetes проверяются ...
    ▪ Используется образ gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Включенные дополнения: storage-provisioner, default-storageclass
🏄  Готово! kubectl настроен для использования кластера "minikube" и "default" пространства имён по умолчанию

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$
```

Статус minikube

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$
```

Включение дополнений

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$ minikube addons enable dashboard
💡  dashboard is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
    ▪ Используется образ docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Используется образ docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🌟  The 'dashboard' addon is enabled

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$ minikube addons enable metrics-server
💡  metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
    ▪ Используется образ registry.k8s.io/metrics-server/metrics-server:v0.8.1
🌟  The 'metrics-server' addon is enabled

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$
```

Проверка создания секрета

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$ kubectl get secrets
NAME          TYPE                             DATA   AGE
ghcr-secret   kubernetes.io/dockerconfigjson   1      63s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-2 (main)
$
```

1.2 Узлы кластера

Информация о кластере

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:58083
CoreDNS is running at https://127.0.0.1:58083/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

Список узлов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   17m   v1.35.1

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide
NAME       STATUS   ROLES           AGE   VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION
    CONTAINER-RUNTIME
minikube   Ready    control-plane   17m   v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   6.6.87.2-microsoft-standard-WSL
2   docker://29.2.1
```

Вывод kubectl describe nodes

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl describe nodes
Name:               minikube
Roles:              control-plane
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    kubernetes.io/arch=amd64
                    kubernetes.io/hostname=minikube
                    kubernetes.io/os=linux
                    minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
                    minikube.k8s.io/name=minikube
                    minikube.k8s.io/primary=true
                    minikube.k8s.io/updated_at=2026_03_21T11_35_51_0700
                    minikube.k8s.io/version=v1.38.1
                    node-role.kubernetes.io/control-plane=
                    node.kubernetes.io/exclude-from-external-load-balancers=
Annotations:        node.alpha.kubernetes.io/ttl: 0
                    volumes.kubernetes.io/controller-managed-attach-detach: true
CreationTimestamp:  Sat, 21 Mar 2026 11:35:47 +0300
Taints:             <none>
Unschedulable:      false
Lease:
  HolderIdentity:  minikube
  AcquireTime:     <unset>
  RenewTime:       Sat, 21 Mar 2026 11:53:01 +0300
Conditions:
  Type             Status  LastHeartbeatTime                 LastTransitionTime                Reason                       Message
  ----             ------  -----------------                 ------------------                ------                       -------
  MemoryPressure   False   Sat, 21 Mar 2026 11:51:46 +0300   Sat, 21 Mar 2026 11:35:46 +0300   KubeletHasSufficientMemory   kubelet has sufficien
t memory available
  DiskPressure     False   Sat, 21 Mar 2026 11:51:46 +0300   Sat, 21 Mar 2026 11:35:46 +0300   KubeletHasNoDiskPressure     kubelet has no disk p
ressure
  PIDPressure      False   Sat, 21 Mar 2026 11:51:46 +0300   Sat, 21 Mar 2026 11:35:46 +0300   KubeletHasSufficientPID      kubelet has sufficien
t PID available
  Ready            True    Sat, 21 Mar 2026 11:51:46 +0300   Sat, 21 Mar 2026 11:35:58 +0300   KubeletReady                 kubelet is posting re
ady status
```

```
Addresses:
  InternalIP:  192.168.49.2
  Hostname:    minikube
Capacity:
  cpu:                12
  ephemeral-storage:  1055762868Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             10183884Ki
  pods:               110
Allocatable:
  cpu:                12
  ephemeral-storage:  1055762868Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             10183884Ki
  pods:               110
System Info:
  Machine ID:                 86e0bfeb3f77427722393c2969964edb
  System UUID:                86e0bfeb3f77427722393c2969964edb
  Boot ID:                    1230f9d5-9bd3-4d5c-b26e-b814948123f5
  Kernel Version:             6.6.87.2-microsoft-standard-WSL2
  OS Image:                   Debian GNU/Linux 12 (bookworm)
  Operating System:           linux
  Architecture:               amd64
  Container Runtime Version:  docker://29.2.1
  Kubelet Version:            v1.35.1
  Kube-Proxy Version:
PodCIDR:                      10.244.0.0/24
PodCIDRs:                     10.244.0.0/24
Non-terminated Pods:          (11 in total)
```

```
  Namespace                   Name                                        CPU Requests  CPU Limits  Memory Requests  Memory Limits  Age
  ---------                   ----                                        ------------  ----------  ---------------  -------------  ---
  kube-system                 coredns-7d764666f9-btjtz                    100m (0%)     0 (0%)      70Mi (0%)        170Mi (1%)     17m
  kube-system                 coredns-7d764666f9-ltklw                    100m (0%)     0 (0%)      70Mi (0%)        170Mi (1%)     17m
  kube-system                 etcd-minikube                               100m (0%)     0 (0%)      100Mi (1%)       0 (0%)         17m
  kube-system                 kube-apiserver-minikube                     250m (2%)     0 (0%)      0 (0%)           0 (0%)         17m
  kube-system                 kube-controller-manager-minikube            200m (1%)     0 (0%)      0 (0%)           0 (0%)         17m
  kube-system                 kube-proxy-v6r7q                            0 (0%)        0 (0%)      0 (0%)           0 (0%)         17m
  kube-system                 kube-scheduler-minikube                     100m (0%)     0 (0%)      0 (0%)           0 (0%)         17m
  kube-system                 metrics-server-9d74bb658-8stvz              100m (0%)     0 (0%)      200Mi (2%)       0 (0%)         14m
  kube-system                 storage-provisioner                         0 (0%)        0 (0%)      0 (0%)           0 (0%)         17m
  kubernetes-dashboard        dashboard-metrics-scraper-5565989548-dwwrf  0 (0%)        0 (0%)      0 (0%)           0 (0%)         14m
  kubernetes-dashboard        kubernetes-dashboard-b84665fb8-b5vmr        0 (0%)        0 (0%)      0 (0%)           0 (0%)         14m
Allocated resources:
  (Total limits may be over 100 percent, i.e., overcommitted.)
  Resource           Requests    Limits
  --------           --------    ------
  cpu                950m (7%)   0 (0%)
  memory             440Mi (4%)  340Mi (3%)
  ephemeral-storage  0 (0%)      0 (0%)
  hugepages-1Gi      0 (0%)      0 (0%)
  hugepages-2Mi      0 (0%)      0 (0%)
Events:
  Type    Reason          Age   From             Message
  ----    ------          ----  ----             -------
  Normal  RegisteredNode  17m   node-controller  Node minikube event: Registered Node minikube in Controller

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

Список доступных API-ресурсов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl api-resources | findstr pod
pods                              po          v1                              true        Pod
podtemplates                                  v1                              true        PodTemplate
horizontalpodautoscalers          hpa         autoscaling/v2                  true        HorizontalPodAutoscaler
pods                                          metrics.k8s.io/v1beta1          true        PodMetrics
poddisruptionbudgets              pdb         policy/v1                       true        PodDisruptionBudget

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl api-resources | findstr deployment
deployments                       deploy      apps/v1                         true        Deployment

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

Просмотр конфигурации

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl config view
apiVersion: v1
clusters:
- cluster:
    certificate-authority: C:\Users\giper\.minikube\ca.crt
    extensions:
    - extension:
        last-update: Sat, 21 Mar 2026 11:35:51 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: cluster_info
    server: https://127.0.0.1:58083
  name: minikube
- cluster:
    certificate-authority-data: DATA+OMITTED
    server: https://158.160.205.16
  name: yc-managed-k8s-catnl1v32nh5fg538k60
contexts:
- context:
    cluster: minikube
    extensions:
    - extension:
        last-update: Sat, 21 Mar 2026 11:35:51 MSK
        provider: minikube.sigs.k8s.io
        version: v1.38.1
      name: context_info
    namespace: default
    user: minikube
  name: minikube
- context:
    cluster: yc-managed-k8s-catnl1v32nh5fg538k60
    user: yc-managed-k8s-catnl1v32nh5fg538k60
  name: yc-test-kubernetes
current-context: minikube
kind: Config
users:
- name: minikube
  user:
    client-certificate: C:\Users\giper\.minikube\profiles\minikube\client.crt
    client-key: C:\Users\giper\.minikube\profiles\minikube\client.key
- name: yc-managed-k8s-catnl1v32nh5fg538k60
  user:
    exec:
      apiVersion: client.authentication.k8s.io/v1beta1
      args:
      - k8s
      - create-token
      - --profile=default
      command: C:\Users\giper\yandex-cloud\bin\yc.exe
      env: null
      interactiveMode: IfAvailable
      provideClusterInfo: false

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl config current-context
minikube

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

Практическое задание

Сохранение вывода команды kubectl get nodes -o wide в файл nodes.txt

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$ kubectl get nodes -o wide > nodes.txt

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1/k8s (lab2/kubernetes-basics)
$
```

```
k8s > nodes.txt
NAME       STATUS   ROLES           AGE   VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                       KERNEL-VERSION                   CONTAINER-RUNTIME
minikube   Ready    control-plane   22m   v1.35.1   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   6.6.87.2-microsoft-standard-WSL2   docker://29.2.1
```

2. Созданные ресурсы

2.1 Pods

Применение манифеста

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/01-pod-nginx.yaml
pod/nginx-pod created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр подов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
nginx-pod   1/1     Running   0          81s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -o wide
NAME        READY   STATUS    RESTARTS   AGE   IP          NODE       NOMINATED NODE   READINESS GATES
nginx-pod   1/1     Running   0          83s   10.244.0.7  minikube   <none>           <none>

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods --show-labels
NAME        READY   STATUS    RESTARTS   AGE   LABELS
nginx-pod   1/1     Running   0          96s   app=nginx,lab=lab2

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Детальная информация

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe pod nginx-pod
Name:             nginx-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sat, 21 Mar 2026 12:50:38 +0300
Labels:           app=nginx
                  lab=lab2
Annotations:      <none>
Status:           Running
IP:               10.244.0.7
IPs:
  IP:  10.244.0.7
Containers:
  nginx:
    Container ID:   docker://e4098ed0d0432f45c9b4309da3ad82202cc2e536f1abf06a84140315003e5da2
    Image:          nginx:alpine
    Image ID:       docker-pullable://nginx@sha256:f46cb72c7df02710e693e863a983ac42f6a9579058a59a35f1ae36c9958e4ce0
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Sat, 21 Mar 2026 12:50:47 +0300
    Ready:          True
    Restart Count:  0
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:        250m
      memory:     64Mi
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-2sm8v (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-2sm8v:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   Burstable
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason     Age    From               Message
  ----    ------     ----   ----               -------
  Normal  Scheduled  2m41s  default-scheduler  Successfully assigned default/nginx-pod to minikube
  Normal  Pulling    2m41s  kubelet            Pulling image "nginx:alpine"
  Normal  Pulled     2m33s  kubelet            Successfully pulled image "nginx:alpine" in 8.134s (8.134s including waiting). Image size: 62174486 bytes.
  Normal  Created    2m32s  kubelet            Container created
  Normal  Started    2m32s  kubelet            Container started
```

Логи

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs nginx-pod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/21 09:50:47 [notice] 1#1: using the "epoll" event method
2026/03/21 09:50:47 [notice] 1#1: nginx/1.29.6
2026/03/21 09:50:47 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
2026/03/21 09:50:47 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/21 09:50:47 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/21 09:50:47 [notice] 1#1: start worker processes
2026/03/21 09:50:47 [notice] 1#1: start worker process 30
2026/03/21 09:50:47 [notice] 1#1: start worker process 31
2026/03/21 09:50:47 [notice] 1#1: start worker process 32
2026/03/21 09:50:47 [notice] 1#1: start worker process 33
2026/03/21 09:50:47 [notice] 1#1: start worker process 34
2026/03/21 09:50:47 [notice] 1#1: start worker process 35
2026/03/21 09:50:47 [notice] 1#1: start worker process 36
2026/03/21 09:50:47 [notice] 1#1: start worker process 37
2026/03/21 09:50:47 [notice] 1#1: start worker process 38
2026/03/21 09:50:47 [notice] 1#1: start worker process 39
2026/03/21 09:50:47 [notice] 1#1: start worker process 40
2026/03/21 09:50:47 [notice] 1#1: start worker process 41

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Выполнение команд в поде

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl exec -it nginx-pod -- sh
/ # curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр метрик

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl top pod nginx-pod
NAME        CPU(cores)   MEMORY(bytes)
nginx-pod   1m           10Mi

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Удаление пода

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl delete pod nginx-pod
pod "nginx-pod" deleted from default namespace

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Практическое задание

Манифест пода

```yaml
k8s > ! 02-pod-go-app.yaml
apiVersion: v1
kind: Pod
metadata:
  name: go-app-pod
  labels:
    app: go
    lab: lab2
spec:
  imagePullSecrets:
    - name: ghcr-secret
  containers:
    - name: go-app
      image: ghcr.io/hyperslava/containers-lab-1/app:sha-0a34ffc
      ports:
        - containerPort: 80
      resources:
        requests:
          memory: "256Mi"
          cpu: "250m"
        limits:
          memory: "512Mi"
          cpu: "500m"
```

Статус пода после запуска

```
Events:
  Type     Reason     Age                From               Message
  ----     ------     ----               ----               -------
  Normal   Scheduled  3m37s              default-scheduler  Successfully assigned default/go-app-pod to minikube
  Normal   Pulled     26s (x6 over 3m37s)  kubelet          Container image "ghcr.io/hyperslava/containers-lab-1/app:sha-0a34ffc" already
 present on machine and can be accessed by the pod
  Normal   Created    25s (x6 over 3m37s)  kubelet          Container created
  Normal   Started    25s (x6 over 3m36s)  kubelet          Container started
  Warning  BackOff    24s (x5 over 3m35s)  kubelet          Back-off restarting failed container go-app in pod go-app-pod_default(8d5fabb
6-8ef7-4000-a6b9-88ce5d5e0162)
```

Сохранение логов в файл

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-pod > crash-logs.txt

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

```
crash-logs.txt
2026/03/22 10:52:59 Error creating table:dial tcp [::1]:5432: connect: connection refused
```

Работа с ReplicaSet

Создание ReplicaSet

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/02-replicaset-nginx.yaml
replicaset.apps/nginx-replicaset created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр ReplicaSet

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get rs
NAME               DESIRED   CURRENT   READY   AGE
nginx-replicaset   3         3         3       75s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe rs nginx-replicaset
Name:         nginx-replicaset
Namespace:    default
Selector:     app=nginx
Labels:       app=nginx
              lab=lab2
Annotations:  <none>
Replicas:     3 current / 3 desired
Pods Status:  3 Running / 0 Waiting / 0 Succeeded / 0 Failed
Pod Template:
  Labels:  app=nginx
  Containers:
   nginx:
    Image:         nginx:alpine
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Events:
  Type    Reason            Age   From                   Message
  ----    ------            ----  ----                   -------
  Normal  SuccessfulCreate  95s   replicaset-controller  Created pod: nginx-replicaset-f4gd2
  Normal  SuccessfulCreate  95s   replicaset-controller  Created pod: nginx-replicaset-4pcz7
  Normal  SuccessfulCreate  95s   replicaset-controller  Created pod: nginx-replicaset-7nxqg

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр созданных подов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx
NAME                     READY   STATUS    RESTARTS   AGE
nginx-replicaset-4pcz7   1/1     Running   0          3m36s
nginx-replicaset-7nxqg   1/1     Running   0          3m36s
nginx-replicaset-f4gd2   1/1     Running   0          3m36s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Масштабирование вручную

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale rs nginx-replicaset --replicas=5
replicaset.apps/nginx-replicaset scaled

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Наблюдение вручную

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx -w
NAME                    READY   STATUS    RESTARTS   AGE
nginx-replicaset-4pcz7  1/1     Running   0          4m43s
nginx-replicaset-6hc6d  1/1     Running   0          29s
nginx-replicaset-7nxqg  1/1     Running   0          4m43s
nginx-replicaset-f4gd2  1/1     Running   0          4m43s
nginx-replicaset-wpsnl  1/1     Running   0          29s
```

Эксперимент с самовосстановлением

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl delete pod nginx-replicaset-4pcz7
pod "nginx-replicaset-4pcz7" deleted from default namespace

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=nginx
NAME                    READY   STATUS    RESTARTS   AGE
nginx-replicaset-6hc6d  1/1     Running   0          2m2s
nginx-replicaset-7nxqg  1/1     Running   0          6m16s
nginx-replicaset-f4gd2  1/1     Running   0          6m16s
nginx-replicaset-vltdk  1/1     Running   0          23s
nginx-replicaset-wpsnl  1/1     Running   0          2m2s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

2.2 Deployments

Применение манифеста

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml
deployment.apps/go-app-deployment created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр Deployment

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get deployments
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
go-app-deployment    0/2     2            0           109s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe deployment go-app-deployment
Name:                   go-app-deployment
Namespace:              default
CreationTimestamp:      Sat, 21 Mar 2026 14:21:42 +0300
Labels:                 app=go-app
                        lab=lab2
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=go-app
Replicas:               2 desired | 2 updated | 2 total | 0 available | 2 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  0 max unavailable, 1 max surge
Pod Template:
  Labels:  app=go-app
  Containers:
   go-app:
    Image:      ghcr.io/hyperslava/containers-lab-1/app:sha-0a34ffc
    Port:       8080/TCP
    Host Port:  0/TCP
    Limits:
      cpu:     500m
      memory:  128Mi
    Requests:
      cpu:      250m
      memory:   64Mi
    Liveness:   http-get http://:8080/api/health delay=15s timeout=1s period=20s #success=1 #failure=3
    Readiness:  http-get http://:8080/api/health delay=5s timeout=1s period=10s #success=1 #failure=3
    Environment:
      DB_HOST:      postgres-service
      DB_PORT:      5432
      DB_USER:      postgres
      DB_PASSWORD:  secret
      DB_NAME:      myapp
    Mounts:         <none>
  Volumes:          <none>
  Node-Selectors:   <none>
  Tolerations:      <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      False   MinimumReplicasUnavailable
  Progressing    True    ReplicaSetUpdated
OldReplicaSets:  <none>
NewReplicaSet:   go-app-deployment-5d8f4f4645 (2/2 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  2m15s  deployment-controller  Scaled up replica set go-app-deployment-5d8f4f4645 from 0 to 2

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр созданного ReplicaSet

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get rs -l app=go-app
NAME                          DESIRED   CURRENT   READY   AGE
go-app-deployment-5d8f4f4645  2         2         0       3m30s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр подов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=go-app
NAME                                  READY   STATUS             RESTARTS      AGE
go-app-deployment-5d8f4f4645-7slj6    0/1     CrashLoopBackOff   5 (48s ago)   4m1s
go-app-deployment-5d8f4f4645-dqs44    0/1     CrashLoopBackOff   5 (49s ago)   4m1s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр истории обновлений

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl rollout history deployment go-app-deployment
deployment.apps/go-app-deployment
REVISION  CHANGE-CAUSE
1         <none>


giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Масштабирование через Deployment

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale deployment go-app-deployment --replicas=3
deployment.apps/go-app-deployment scaled

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Обновление образа

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl set image deployment/go-app-deployment go-app=ghcr.io/hyperslava/containers-lab-1/app:new-tag
deployment.apps/go-app-deployment image updated

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Откат изменений

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl rollout undo deployment go-app-deployment
deployment.apps/go-app-deployment rolled back

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Практическое задание

Deployment для PostgreSQL

k8s > ! 03-deployment-postgresql.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: postgres-deployment
  labels:
    app: postgres
    lab: lab2
spec:
  replicas: 2
  selector:
    matchLabels:
      app: postgres
  # Стратегия обновления
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  # Шаблон пода
  template:
    metadata:
      labels:
        app: postgres
    spec:
      containers:
      - name: postgres
        image: postgres:alpine
        ports:
        - containerPort: 5432
        env:
        - name: POSTGRES_DB
          value: "myapp"
        - name: POSTGRES_USER
          value: "postgres"
        - name: POSTGRES_PASSWORD
          value: "secret"
        resources:
          requests:
            memory: "64Mi"
            cpu: "250m"
          limits:
            memory: "128Mi"
            cpu: "500m"
        # Проверки готовности и жизнеспособности
        readinessProbe:
          exec:
            command:
              - sh
              - -c
              - pg_isready -U "$POSTGRES_USER" -d "$POSTGRES_DB"
          initialDelaySeconds: 10
          periodSeconds: 5
          timeoutSeconds: 3
          failureThreshold: 6

```

```yaml
          livenessProbe:
            exec:
              command:
                - sh
                - -c
                - pg_isready -U "$POSTGRES_USER" -d "$POSTGRES_DB"
            initialDelaySeconds: 30
            periodSeconds: 10
            timeoutSeconds: 5
            failureThreshold: 6
```

Применение манифеста

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-postgresql.yaml
deployment.apps/postgres-deployment created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Статус Deployments

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
go-app-deployment     0/3     3            0           31m
postgres-deployment   2/2     2            2           3m44s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Статус подов и логи Go-App

```
NAME                                  READY   STATUS             RESTARTS        AGE
go-app-deployment-5d8f4f4645-56rsz    0/1     CrashLoopBackOff   10 (9s ago)     27m
go-app-deployment-5d8f4f4645-7slj6    0/1     CrashLoopBackOff   11 (10s ago)    33m
go-app-deployment-5d8f4f4645-dqs44    0/1     CrashLoopBackOff   11 (29s ago)    33m
go-app-deployment-99dfddc8d-thxlr     0/1     ImagePullBackOff   0               26m
go-app-pod                            0/1     CrashLoopBackOff   21 (2m ago)     86m
nginx-replicaset-6hc6d                1/1     Running            0               71m
nginx-replicaset-7nxqg                1/1     Running            0               75m
nginx-replicaset-f4gd2                1/1     Running            0               75m
nginx-replicaset-vltdk                1/1     Running            0               70m
nginx-replicaset-wpsnl                1/1     Running            0               71m
postgres-deployment-57cb6d54f9-fpqgh  1/1     Running            0               5m24s
postgres-deployment-57cb6d54f9-hjstf  1/1     Running            0               5m24s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-deployment-5d8f4f4645-56rsz
2026/03/21 11:54:33 Error creating table:dial tcp: lookup postgres-service on 10.96.0.10:53: server misbehaving

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

2.3 Services

Создание сервисов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/04-service-postgres.yaml
service/postgres-service created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/05-service-app.yaml
service/go-app-service created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/06-service-nginx.yaml
service/nginx-service created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр сервисов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
go-app-service     ClusterIP   10.108.219.129   <none>        8080/TCP       4m9s
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        3h41m
nginx-service      NodePort    10.98.101.164    <none>        80:30080/TCP   3m55s
postgres-service   ClusterIP   10.102.134.230   <none>        5432/TCP       4m17s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Детальная информация

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl describe svc postgres-service
Name:                     postgres-service
Namespace:                default
Labels:                   app=postgres
                          lab=lab2
Annotations:              <none>
Selector:                 app=postgres
Type:                     ClusterIP
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.102.134.230
IPs:                      10.102.134.230
Port:                     <unset>  5432/TCP
TargetPort:               5432/TCP
Endpoints:                10.244.0.29:5432,10.244.0.28:5432
Session Affinity:         None
Internal Traffic Policy:  Cluster
Events:                   <none>

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Проверка DNS-резолвинга внутри кластера

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run test-pod --image=alpine:latest --rm -it --restart=Never -- sh
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # apk add curl postgresql-client
[fetch https://dl-cdn.alpinelinux.org/alpine/v3.23/main/x86_64/APKINDEX.tar.gz]
[fetch https://dl-cdn.alpinelinux.org/alpine/v3.23/community/x86_64/APKINDEX.tar.gz]
( 1/17) Installing brotli-libs (1.2.0-r0)
( 2/17) Installing c-ares (1.34.6-r0)
( 3/17) Installing libunistring (1.4.1-r0)
( 4/17) Installing libidn2 (2.3.8-r0)
( 5/17) Installing nghttp2-libs (1.68.0-r0)
( 6/17) Installing nghttp3 (1.13.1-r0)
( 7/17) Installing libpsl (0.21.5-r3)
( 8/17) Installing zstd-libs (1.5.7-r2)
( 9/17) Installing libcurl (8.17.0-r1)
(10/17) Installing curl (8.17.0-r1)
(11/17) Installing postgresql-common (1.2-r2)
  Executing postgresql-common-1.2-r2.pre-install
(12/17) Installing lz4-libs (1.10.0-r0)
(13/17) Installing libpq (18.3-r0)
(14/17) Installing ncurses-terminfo-base (6.5_p20251123-r0)
(15/17) Installing libncursesw (6.5_p20251123-r0)
(16/17) Installing readline (8.3.1-r0)
(17/17) Installing postgresql18-client (18.3-r0)
Executing busybox-1.37.0-r30.trigger
Executing postgresql-common-1.2-r2.trigger
* Setting postgresql18 as the default version
OK: 17.6 MiB in 33 packages
/ #
/ #
/ # curl go-app-service:8080/api/health
{"status":"healthy"}
/ # pg_isready -h postgres-service
postgres-service:5432 - accepting connections
/ # exit
pod "test-pod" deleted from default namespace

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Валидация YAML

Проверка синтаксиса YAML с помощью онлайн-сервиса yamllint

https://www.yamllint.com

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

```yaml
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: postgres-deployment
  labels:
    app: postgres
    lab: lab2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: postgres
  template:
    metadata:
      labels:
        app: postgres
    spec:
      containers:
        - name: postgres
```

Go ☑ Reformat (strips comments) ☑ Resolve aliases

Multiple documents detected - Valid YAML!

Сухая проверка (dry-run) в Kubernetes

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/03-deployment-app.yaml --dry-run=client
deployment.apps/go-app-deployment configured (dry run)
```

Получение объяснения по полям

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain pod
KIND:       Pod
VERSION:    v1

DESCRIPTION:
    Pod is a collection of containers that can run on a host. This resource is
    created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind  <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata      <ObjectMeta>
    Standard object's metadata. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#metadata

  spec  <PodSpec>
    Specification of the desired behavior of the pod. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status

  status        <PodStatus>
    Most recently observed status of the pod. This data may not be up to date.
    Populated by the system. Read-only. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#spec-and-status
```

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: spec <DeploymentSpec>


DESCRIPTION:
    Specification of the desired behavior of the Deployment.
    DeploymentSpec is the specification of the desired behavior of the
    Deployment.

FIELDS:
  minReadySeconds       <integer>
    Minimum number of seconds for which a newly created pod should be ready
    without any of its container crashing, for it to be considered available.
    Defaults to 0 (pod will be considered available as soon as it is ready)

  paused        <boolean>
    Indicates that the deployment is paused.

  progressDeadlineSeconds       <integer>
    The maximum time in seconds for a deployment to make progress before it is
    considered to be failed. The deployment controller will continue to process
    failed deployments and a condition with a ProgressDeadlineExceeded reason
    will be surfaced in the deployment status. Note that progress will not be
    estimated during the time a deployment is paused. Defaults to 600s.

  replicas      <integer>
    Number of desired pods. This is a pointer to distinguish between explicit
    zero and not specified. Defaults to 1.
```

```
   revisionHistoryLimit  <integer>
     The number of old ReplicaSets to retain to allow rollback. This is a pointer
     to distinguish between explicit zero and not specified. Defaults to 10.

   selector      <LabelSelector> -required-
     Label selector for pods. Existing ReplicaSets whose pods are selected by
     this will be the ones affected by this deployment. It must match the pod
     template's labels.

   strategy      <DeploymentStrategy>
     The deployment strategy to use to replace existing pods with new ones.

   template      <PodTemplateSpec> -required-
     Template describes the pods that will be created. The only allowed
     template.spec.restartPolicy value is "Always".



giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl explain deployment.spec.template.spec.containers
GROUP:      apps
KIND:       Deployment
VERSION:    v1

FIELD: containers <[]Container>


DESCRIPTION:
    List of containers belonging to the pod. Containers cannot currently be
    added or removed. There must be at least one container in a Pod. Cannot be
    updated.
    A single application container that you want to run within a pod.

FIELDS:
  args  <[]string>
    Arguments to the entrypoint. The container image's CMD is used if this is
    not provided. Variable references $(VAR_NAME) are expanded using the
    container's environment. If a variable cannot be resolved, the reference in
    the input string will be unchanged. Double $$ are reduced to a single $,
    which allows for escaping the $(VAR_NAME) syntax: i.e. "$$(VAR_NAME)" will
    produce the string literal "$(VAR_NAME)". Escaped references will never be
    expanded, regardless of whether the variable exists or not. Cannot be
    updated. More info:
    https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell

  command       <[]string>
    Entrypoint array. Not executed within a shell. The container image's
    ENTRYPOINT is used if this is not provided. Variable references $(VAR_NAME)
    are expanded using the container's environment. If a variable cannot be
    resolved, the reference in the input string will be unchanged. Double $$ are
    reduced to a single $, which allows for escaping the $(VAR_NAME) syntax:
    i.e. "$$(VAR_NAME)" will produce the string literal "$(VAR_NAME)". Escaped
    references will never be expanded, regardless of whether the variable exists
    or not. Cannot be updated. More info:
    https://kubernetes.io/docs/tasks/inject-data-application/define-command-argument-container/#running-a-command-in-a-shell
```

Валидация с помощью kubeconform (это усовершенствованная версия kubeval, который устарел)

```
C:\Dev\containers-lab-1>kubeconform -v
v0.7.0

C:\Dev\containers-lab-1>kubeconform k8s/03-deployment-app.yaml

C:\Dev\containers-lab-1>kubeconform k8s/04-service-postgres.yaml

C:\Dev\containers-lab-1>kubeconform k8s/05-service-app.yaml

C:\Dev\containers-lab-1>kubeconform k8s/06-service-nginx.yaml

C:\Dev\containers-lab-1>kubeconform k8s/07-postgres-deployment.yaml

C:\Dev\containers-lab-1>kubeconform k8s/08-nginx-deployment.yaml

C:\Dev\containers-lab-1>
```

Отладка приложений

Просмотр событий в кластере

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get events --sort-by='.lastTimestamp'
LAST SEEN   TYPE      REASON    OBJECT           MESSAGE
4m57s       Warning   BackOff   pod/go-app-pod   Back-off restarting failed container go-app in pod go-app-pod_default(8d5fabb6-8ef7-4000-a6b9-88ce5d5e0162)
3m21s       Normal    Pulled    pod/go-app-pod   Container image "ghcr.io/hyperslava/containers-lab-1/app:sha-0a34ffc" already present on machine and can be acces
sed by the pod

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр логов конкретного пода

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs -l app=go-app
2026/03/21 16:36:28 Server starting on port 8080
2026/03/21 16:36:27 Server starting on port 8080
2026/03/21 16:36:28 Server starting on port 8080

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Просмотр логов предыдущей инстанции пода

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs go-app-deployment-5d8f4f4645-56rsz --previous
2026/03/21 16:36:15 Error creating table:dial tcp: lookup postgres-service on 10.96.0.10:53: read udp 10.244.0.35:52043->10.96.0.10:53: i/o timeout

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Порт-форвардинг для доступа к приложению без Service

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl port-forward pod/go-app-deployment-5d8f4f4645-56rsz 8080:8080
Forwarding from 127.0.0.1:8080 -> 8080
Forwarding from [::1]:8080 -> 8080
```

Интерактивная отладка - запуск временного пода в той же сети

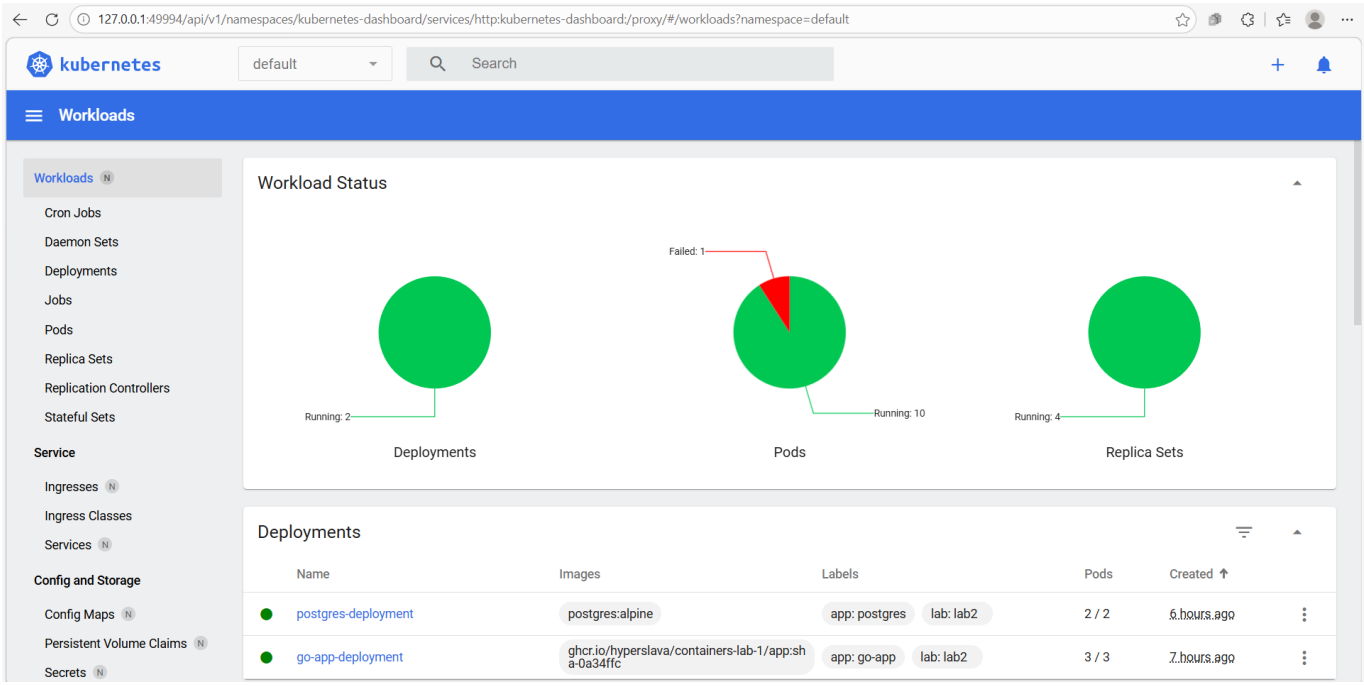
```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl run debug --image=nicolaka/netshoot:latest -it --rm -- bash
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
debug:~# curl go-app-service:8080/api/health
{"status":"healthy"}
debug:~# nslookup postgres-service
;; Got recursion not available from 10.96.0.10
Server:         10.96.0.10
Address:        10.96.0.10#53

Name:   postgres-service.default.svc.cluster.local
Address: 10.102.134.230
;; Got recursion not available from 10.96.0.10

debug:~#
```

Мониторинг через Dashboard

Запуск дашборда

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ minikube dashboard
🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:49994/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
```

Открытый дашборд (1 упавший под - из практического задания в 2.1 Pods)



Последовательный запуск всех компонентов

Применение всех манифестов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl apply -f k8s/
pod/nginx-pod created
pod/go-app-pod unchanged
replicaset.apps/nginx-replicaset configured
deployment.apps/go-app-deployment configured
service/postgres-service unchanged
service/go-app-service unchanged
service/nginx-service unchanged
deployment.apps/postgres-deployment created
configmap/postgres-init-sql created
deployment.apps/nginx-deployment created
configmap/nginx-config created
configmap/nginx-html created

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Проверка статуса

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get all
NAME                                      READY   STATUS             RESTARTS        AGE
pod/go-app-deployment-5d8f4f4645-56rsz    1/1     Running            19 (56s ago)    7h18m
pod/go-app-deployment-5d8f4f4645-7slj6    1/1     Running            20 (59s ago)    7h24m
pod/go-app-pod                            0/1     CrashLoopBackOff   59 (3m4s ago)   8h
pod/nginx-deployment-6f7bb479f4-42dtp     1/1     Running            0               63s
pod/nginx-deployment-6f7bb479f4-qjhlx     1/1     Running            0               61s
pod/postgres-deployment-75fc8ccd7-xh69j   1/1     Running            0               63s

NAME                      TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/go-app-service    ClusterIP   10.108.219.129   <none>        8080/TCP       6h32m
service/kubernetes        ClusterIP   10.96.0.1        <none>        443/TCP        10h
service/nginx-service     NodePort    10.98.101.164    <none>        80:30080/TCP   6h32m
service/postgres-service  ClusterIP   10.102.134.230   <none>        5432/TCP       6h32m

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/go-app-deployment     2/2     2            2           7h24m
deployment.apps/nginx-deployment      2/2     2            2           63s
deployment.apps/postgres-deployment   1/1     1            1           63s

NAME                                            DESIRED   CURRENT   READY   AGE
replicaset.apps/go-app-deployment-5d8f4f4645    2         2         2       7h24m
replicaset.apps/go-app-deployment-99dfddc8d     0         0         0       7h18m
replicaset.apps/nginx-deployment-6f7bb479f4     2         2         2       63s
replicaset.apps/nginx-replicaset                0         0         0       8h
replicaset.apps/postgres-deployment-75fc8ccd7   1         1         1       63s

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Ожидание готовности

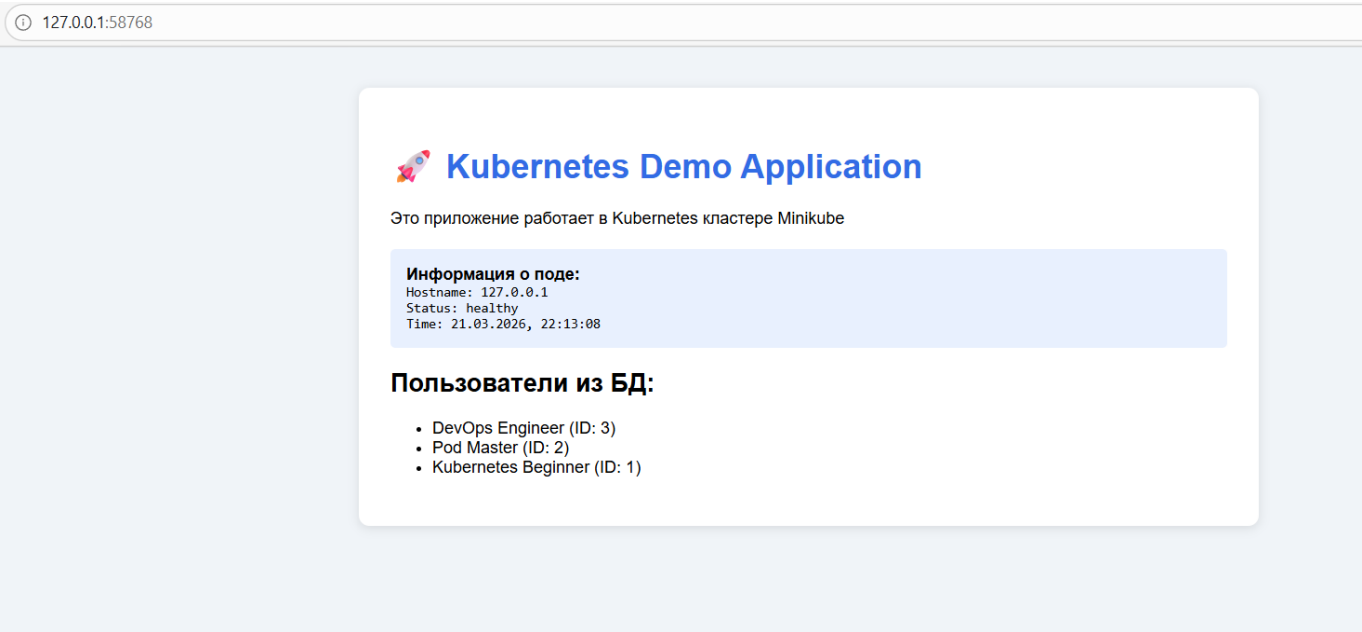
```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=postgres --timeout=60s
pod/postgres-deployment-75fc8ccd7-xh69j condition met

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=go-app --timeout=60s
pod/go-app-deployment-5d8f4f4645-56rsz condition met
pod/go-app-deployment-5d8f4f4645-7slj6 condition met

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl wait --for=condition=ready pod -l app=nginx --timeout=60s
pod/nginx-deployment-6f7bb479f4-42dtp condition met
pod/nginx-deployment-6f7bb479f4-qjhlx condition met

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Получение URL для доступа к приложению

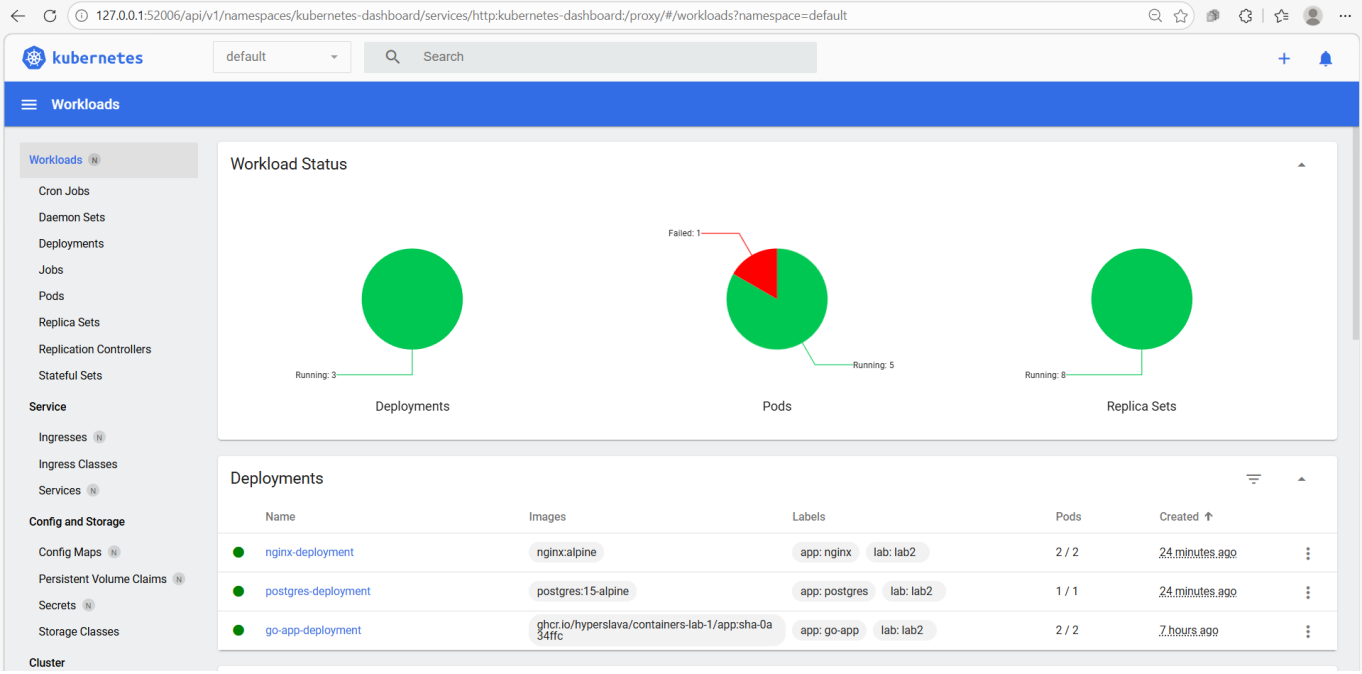
```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ minikube service nginx-service --url
http://127.0.0.1:58768
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

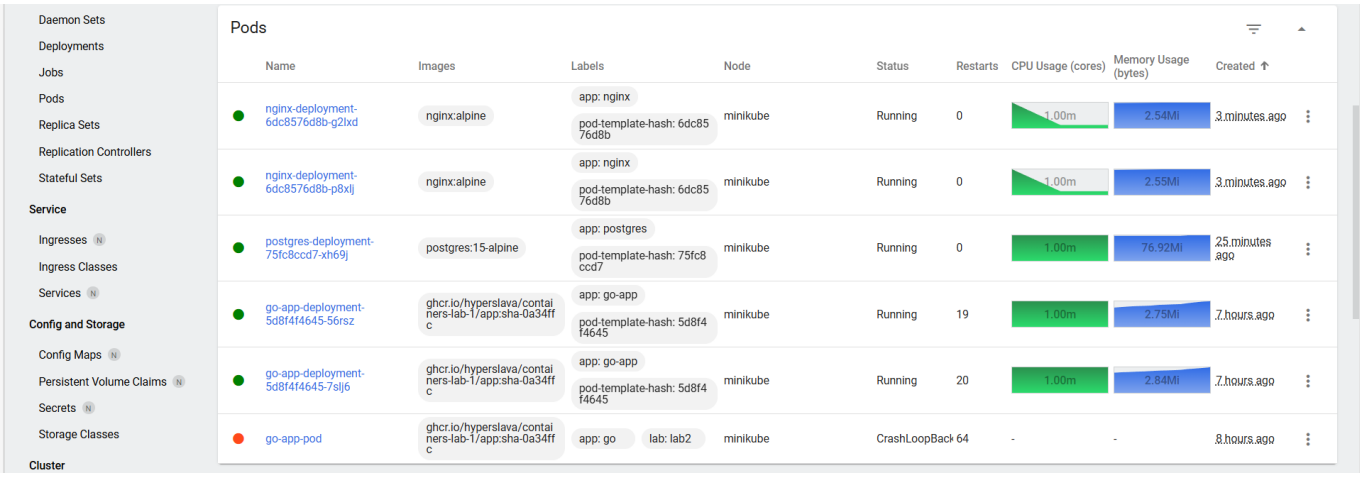
3. Скриншоты работы приложения

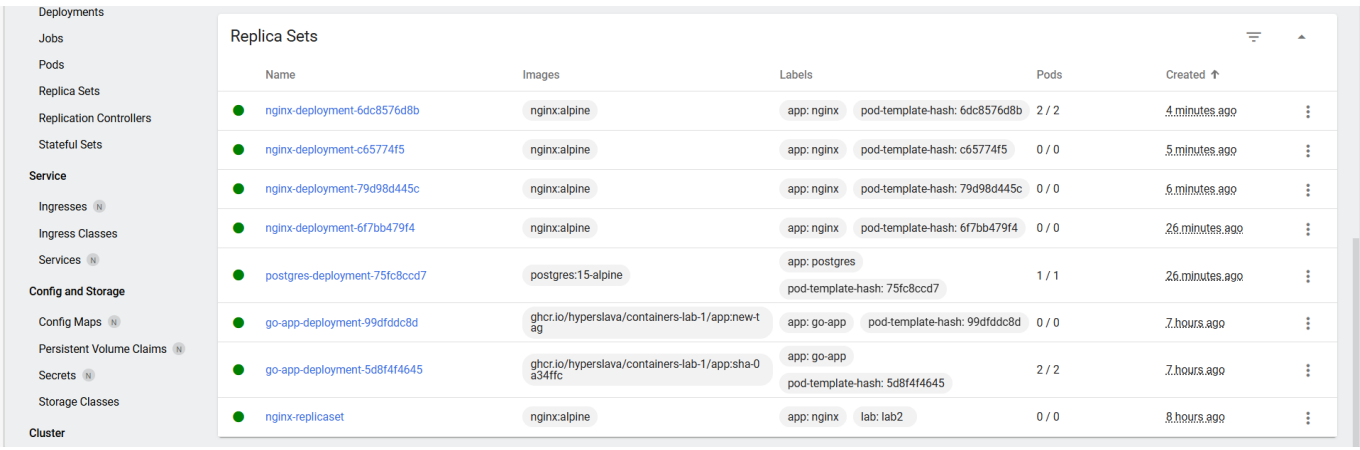
3.1 Главная страница



3.2 Дашборд Kubernetes







3.3 Результат GET /api/users

127.0.0.1:58768/api/users

Качественная печать


```
[
  {
    "id": 3,
    "name": "DevOps Engineer"
  },
  {
    "id": 2,
    "name": "Pod Master"
  },
  {
    "id": 1,
    "name": "Kubernetes Beginner"
  }
]
```


4. Эксперименты с масштабированием

4.1 Масштабирование до 5 реплик

Масштабирование до 5 реплик

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl scale deployment go-app-deployment --replicas=5
deployment.apps/go-app-deployment scaled

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

Статус подов Go-App после масштабирования

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl get pods -l app=go-app
NAME                                READY   STATUS    RESTARTS        AGE
go-app-deployment-5d8f4f4645-4l5pp  1/1     Running   0               11m
go-app-deployment-5d8f4f4645-56rsz  1/1     Running   19 (41m ago)    7h59m
go-app-deployment-5d8f4f4645-6vf2g  1/1     Running   0               11m
go-app-deployment-5d8f4f4645-7slj6  1/1     Running   20 (41m ago)    8h
go-app-deployment-5d8f4f4645-hnshb  1/1     Running   0               11m

giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$
```

4.2 Проверка распределения нагрузки

Команда PowerShell для нагрузочного тестирования

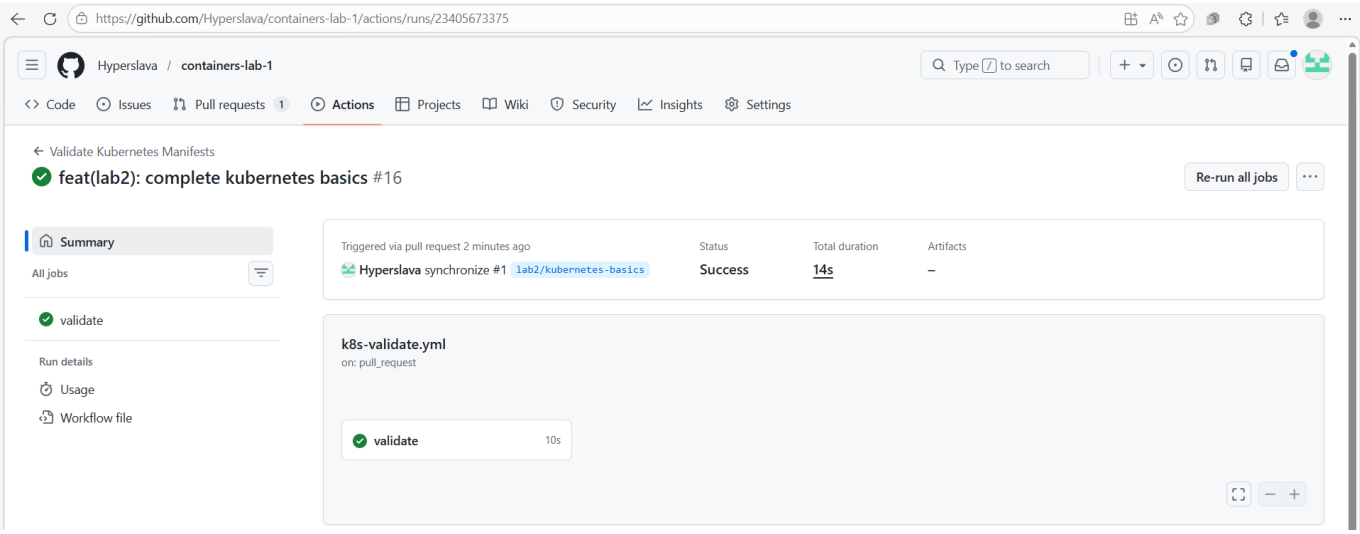
```
PS C:\Dev\containers-lab-1> for ($i=1; $i -le 100; $i++) {
>>   curl http://localhost:58768/api/users
>> Start-Sleep -Milliseconds 100
>> }
```

Логи nginx с разных подов

```
giper@LAPTOP-31PBTHTU MINGW64 /c/Dev/containers-lab-1 (lab2/kubernetes-basics)
$ kubectl logs -l app=nginx --tail=10 -f --prefix
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:07 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:07 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:07 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:07 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:07 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:08 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:08 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-g2lxd/nginx] 10.244.0.1 - - [21/Mar/2026:19:27:08 +0000] "GET /api/users HTTP/1.1" 200 103 "-" "Mozilla/5.0 (Windows NT; Wi
ndows NT 10.0; ru-RU) WindowsPowerShell/5.1.26100.7920"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:10 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:58768/" "Mozil
la/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:10 +0000] "GET /api/users HTTP/1.1" 200 103 "http://127.0.0.1:58768/" "Mozil
la/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:10 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://127.0.0.1:58768/" "Moz
illa/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 2026/03/21 19:13:10 [error] 30#30: *17 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or d
irectory), client: 10.244.0.1, server: , request: "GET /favicon.ico HTTP/1.1", host: "127.0.0.1:58768", referrer: "http://127.0.0.1:58768/"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:14 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:58768/" "Mozil
la/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:19 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:58768/" "Mozil
la/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
[pod/nginx-deployment-6dc8576d8b-p8xlj/nginx] 10.244.0.1 - - [21/Mar/2026:19:13:24 +0000] "GET /api/health HTTP/1.1" 200 21 "http://127.0.0.1:58768/" "Mozil
la/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 Edg/146.0.0.0"
```

5. GitHub Actions

5.1 Успешная валидация манифестов



6. Ответы на контрольные вопросы

1. В чем разница между Pod и Deployment?

Pod — минимальная единица развертывания в Kubernetes, Deployment — более высокоуровневый объект, который управляет подами, ReplicaSet и обеспечивает стратегии обновления, отката и масштабирования.

2. Для чего нужен Service типа ClusterIP?

Для того, чтобы сервис был доступен только внутри кластера.

3. Как ReplicaSet обеспечивает самовосстановление?

При удалении какого-то пода из ReplicaSet взамен него автоматически создаётся новый.

4. Что произойдет с приложением, если удалить под PostgreSQL?

Если удалить под PostgreSQL, то приложение продолжит работать, на странице в информации о поде в поле Status значение поменяется на unhealthy, пропадут записи о пользователях из БД. Затем через небольшой промежуток времени автоматически создадится новый под PostgreSQL, в поле Status снова будет записано healthy, и записи о пользователях снова появятся, они подтянутся из init-script.

7. Выводы

Из нового я узнал, как работать с Minikube, валидировать YAML, проводить отладку приложений, проводить проверку DNS-резолвинга внутри кластера, мониторинг через Dashboard и как проводить нагрузочное тестирование. С трудностями не столкнулся.